TinkerWithMe

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AI Game Designer · Full Curriculum

Full-day camp (5 hrs) · 10–14 yrs · Intermediate

Full facilitator curriculum PREMIUM

Overview

Design and build a playable game using AI tools.

DETAIL	
Track	AI Skills for Kids
Format	Full-day camp (5 hrs)
Ages	10–14 yrs
Level	Intermediate
Price	from KES 7,000
Standalone	No
Prerequisites	Prompting power (free, recommended)

Course Overview

AI Game Designer is a full-day intensive in which learners act as a complete game studio-of-one: designer, writer, illustrator, sound engineer and developer — with generative AI as their creative co-pilot. Working individually or in pairs, students move from a blank page to a fully playable digital game in a single day, while building a professional-style design portfolio that documents every decision along the way.

The day has a clear narrative arc. The **morning (Design)** session is all about ideas made real: students use ChatGPT to brainstorm original game concepts, write formal rules, and flesh out character backstories, then use DALL·E-powered image generation to turn those characters and boards into visual art. The **afternoon (Build)** session shifts from paper to prototype: students bring their designs into Scratch, add AI-generated sound effects to bring the game to life, and then run structured playtesting rounds with peers, using feedback to iterate before the final showcase.

By 3pm every student leaves with two tangible artefacts: a **playable Scratch game** (shareable via a Scratch project link or exported file) and a **design portfolio PDF** containing their concept pitch, rules, character bios, AI-generated artwork, and a short reflection on what they changed after playtesting. This mirrors the real indie-game-design pipeline and gives students a genuine "I made this" artefact to show family, friends, and (for the older end of the age range) future portfolios.

Learning Outcomes

- Use large language model prompting (ChatGPT) to generate, refine and select original game concepts, rules and narrative content
- Write clear, playable game rules including setup, turn structure, win/lose conditions and scoring
- Use text-to-image AI (DALL·E / Bing Image Creator) to generate consistent character art and game board/card designs, including iterating on prompts to improve results
- Build a working digital game in Scratch, including sprite movement, scoring variables, collision detection and win/lose logic
- Generate and integrate AI-created sound effects into a Scratch project
- Run a structured peer playtest, collect actionable feedback, and make specific design iterations in response
- Assemble a professional design portfolio that documents the full creative process from concept to finished game

Materials Master List

ITEM	SPEC	QTY PER STUDENT/GROUP	NOTES
Laptop or Chromebook	Able to run Chrome/Edge browser, keyboard + trackpad/ mouse	1 per student (pair-share acceptable for younger groups)	Pre-install Scratch offline editor as internet backup; disable auto-updates before session

Headphones with mic	Basic wired stereo headset	1 per student	Needed for sound effect review and to avoid audio clash between stations
Reliable internet connection	Minimum 20 Mbps shared, Wi-Fi router	1 per venue	Have a mobile hotspot (Safaricom/Airtel) as backup — image generation is bandwidth-heavy
Power strip/ extension cables	4–6 gang, surge protected	1 per 6 students	Nairobi power cuts: have one power bank per table as backup
Free ChatGPT account	chat.openai.com free tier (GPT-3.5/4o-mini access)	1 per student	Pre-create facilitator "backup" accounts in case of sign-up issues (age gate: 13+; under-13 pairs use facilitator/parent login)
Microsoft account for Bing Image Creator	Free account at bing.com/ images/create (DALL·E 3 engine)	1 per student	Pre-create a shared classroom Microsoft account as fallback for quick sign-up failures
Scratch account	scratch.mit.edu free account, or offline Scratch 3 editor installed	1 per student	Create accounts in advance if possible to save time
ElevenLabs free account	elevenlabs.io free tier — Sound Effects generator	1 per pair	Free tier has a monthly credit limit — pair students to conserve credits
Printed Game Concept Canvas	A4, one-page worksheet (title, genre, theme, core mechanic, win condition)	1 per student	Facilitator-designed template, printed double-sided with Character Bio Sheet on reverse
Printed Character Bio Sheet	A4 worksheet (name, role, strength, flaw, motivation, catchphrase)	1–2 per student	See above — reverse of Concept Canvas
Printed AI Prompt Cheat-Sheet	A5 laminated card with example prompt formulas for ChatGPT, Bing Image Creator, ElevenLabs	1 per student	Laminate for reuse across camps
Sticky notes	76x76mm, assorted colours	1 pad per group of 4	Used in playtest feedback round
Index cards	plain 10x15cm	10 per student	For quick brainstorm sorting and playtest feedback capture
USB flash drives	8GB+	1 per 4 students	Backup export of Scratch .sb3 files and portfolio images

Portfolio folder (digital)	Google Slides or Canva template, pre-shared	1 per student	Facilitator builds and shares a template with placeholder slides in advance
Projector/screen + HDMI/adapter	Any classroom projector	1 per venue	For demos, showcase and example games
Timer (visible)	Large classroom timer or phone projected	1 per venue	Keeps brainstorm and playtest rounds on pace
Name tags/table tents	Card + marker	1 per student	Helps facilitator personalise AI prompts and peer feedback

Session Plans

Morning — Design (Brainstorm, Write, Illustrate)

Objectives

- Generate and select an original game concept using ChatGPT-assisted brainstorming
- Write a complete, playable ruleset and character backstories with AI support
- Produce consistent character art and board/card designs using DALL·E-powered image generation

Timing (150 minutes)

- 0:00–0:10 — Welcome, icebreaker, show 2 example finished games (video or live demo)
- 0:10–0:20 — Introduce the AI Game Designer workflow and the portfolio students will build all day
- 0:20–0:50 — ChatGPT brainstorming: generate and shortlist game concepts (30 min)
- 0:50–1:15 — Write full rules with ChatGPT, refine into Game Concept Canvas (25 min)
- 1:15–1:35 — Generate character backstories with ChatGPT (20 min)
- 1:35–1:45 — Break (10 min)
- 1:45–2:15 — Character art + board/card design with Bing Image Creator (30 min)
- 2:15–2:25 — Paste chosen text and images into portfolio slides (10 min)

- 2:25–2:30 — Recap and preview of afternoon Build session (5 min)

Step-by-step

1. Open with a 2-minute demo of a simple existing Scratch game (e.g. a maze-collector) so students can visualise the end goal before they design.
2. Explain the day's pipeline on the whiteboard: Concept → Rules → Characters → Art → Build → Sound → Playtest → Showcase. Tell students every box will live in their portfolio.
3. Hand out the Game Concept Canvas and AI Prompt Cheat-Sheet. Log everyone into ChatGPT (facilitator walks the room checking sign-ins).
4. Model the first ChatGPT brainstorm prompt live on the projector so students see a real output before trying it themselves.
5. Students run their own brainstorm prompt, read the 5 concepts returned, and circle/star their favourite — facilitator checks in with each student/pair to sanity-check their pick is buildable in Scratch (movement + scoring + win condition, not an open-world RPG).
6. Students ask ChatGPT to expand their chosen concept into full rules, then hand-copy/paste the trimmed rules onto their Game Concept Canvas in their own words (this forces comprehension, not blind copy-paste).
7. Introduce character backstory prompting; each student generates 1–2 characters (protagonist + one obstacle/villain/NPC) and fills the Character Bio Sheet.
8. After the break, introduce Bing Image Creator. Model one prompt live, showing how changing style keywords changes the result.
9. Students generate: (a) one hero character portrait, (b) one board/card/background design consistent with their theme. They save/download best results.
10. Students paste their favourite text and images into their pre-shared portfolio template (Slide 1: Concept, Slide 2: Rules, Slide 3: Characters, Slide 4: Art).
11. Close the session by asking 2–3 students to briefly share their concept aloud to the group, building excitement for the build phase.

Build detail: AI tools, workflow, and exact prompts

Tool setup: ChatGPT at chat.openai.com (free tier is sufficient — no image generation needed here, image generation happens in Bing Image Creator). Bing

Image Creator at bing.com/images/create, signed in with a Microsoft account, uses DALL·E 3. Both should be opened in separate browser tabs before the session starts to save time.

Step 1 — Concept brainstorm (ChatGPT). Example prompt to type/paste:

```
I'm designing a simple game for a Scratch project, for players aged 10-14.
Give me 5 original game concepts that combine the theme "Kenyan wildlife
conservation" with the genre "collector/arcade game."
For each concept give me:
1. Title
2. One-line pitch
3. Core mechanic (what the player actually does each turn/second)
4. Win condition
Keep each concept simple enough to build with sprites, movement, and a score
counter.
```

Good result looks like: 5 distinct, short, concrete concepts (e.g. "Ranger Run — dodge poachers' traps while collecting snares to disarm, win at 20 snares collected"). Facilitator should reject/ask ChatGPT to simplify any concept requiring multiplayer networking, 3D, or complex branching stories.

Step 2 — Rules (ChatGPT). Example prompt:

```
Write complete rules for my game "Ranger Run" for players aged 10-14.
Include: setup, controls/turn structure, scoring, win condition, and lose
condition. Keep it under 200 words and use simple, clear language.
```

Students copy the response into the Concept Canvas but must rewrite at least the win/lose condition in their own words to check understanding.

Step 3 — Character backstory (ChatGPT). Example prompt:

```
Create a 100-word backstory for a character named Zawadi, a young wildlife
ranger in the game "Ranger Run". Include her motivation, one strength, one
flaw, and a short catchphrase she says when she collects a snare.
```

Good result: a short, vivid paragraph with a clear motivation and a usable catchphrase (useful later for a text bubble or sound effect line in Scratch).

Step 4 — Character art (Bing Image Creator). Example prompt:

```
Full-body character design of a young female wildlife ranger, determined
expression, khaki uniform, holding a walkie-talkie, flat vector illustration
style, bright colours, simple shapes, plain white background, mobile game
character art
```

Good result: a single clear character on a plain background that can be cropped and reused as a Scratch sprite. If the image is too cluttered or has extra limbs/objects, coach students to add "simple," "flat design," or "single character, no background" to the prompt and regenerate (Bing gives 4 options — pick the best, don't just accept the first).

Step 5 — Board/card design (Bing Image Creator). Example prompt:

```
Top-down game background, savanna landscape with acacia trees, dirt path,
watering hole, flat vector illustration style, bright colours, designed as a
side-scrolling game background, no characters
```

Testing/iteration: students should generate at least 2 versions of each image and choose the one that best matches their game's mood — this is an explicit design decision they must note in their portfolio ("I chose version 2 because...").

Checks for understanding

- "What is the win condition of your game, in your own words?"
- "Why did you choose this concept over the other 4 ChatGPT gave you?"
- "What word did you change in your image prompt to get a cleaner character, and what changed in the result?"

Common mistakes & fixes

MISTAKE	FIX
Student picks an overly complex concept (multiplayer, branching story, 3D)	Facilitator asks ChatGPT live to "simplify this concept so it only needs sprite movement and a score counter"
Student copy-pastes AI rules without understanding them	Require students to explain the win/lose condition aloud or rewrite it by hand before moving on
Image prompt returns cluttered or distorted art	Coach adding style + simplicity keywords ("flat vector," "single object," "plain background") and regenerating
Character art has a background that clashes with the game	Add "plain white background" or "transparent background" to the prompt so it's easy to crop in Scratch

Afternoon — Build (Scratch, AI Sound, Playtest & Iterate)

Objectives

- Build a working game prototype in Scratch using their morning design (sprites, movement, scoring, win/lose logic)

- Generate and integrate AI sound effects that match the game's theme
- Run a structured playtest with peers and make at least one concrete iteration based on feedback

Timing (150 minutes)

- 0:00–0:10 — Recap morning designs, set the afternoon goal ("playable by 2:30")
- 0:10–0:20 — Scratch tour: interface, sprites, costumes, sounds, blocks palette
- 0:20–0:55 — Core build block: import art, movement, scoring (35 min)
- 0:55–1:20 — AI sound effect generation with ElevenLabs (25 min)
- 1:20–1:30 — Import sounds into Scratch and attach to events (10 min)
- 1:30–1:40 — Break (10 min)
- 1:40–1:55 — Polish: backgrounds, win/lose screens, score display (15 min)
- 1:55–2:10 — Structured peer playtesting rotation (15 min)
- 2:10–2:20 — Iterate based on feedback (10 min)
- 2:20–2:30 — Final showcase / gallery walk and wrap-up (10 min)

Step-by-step

1. Recap each student's concept aloud in one sentence to re-energise the room and remind everyone what they're building.
2. Give a live 10-minute Scratch tour on the projector: stage, sprite list, costume tab, sound tab, and the 3 block categories students will use most (Motion, Looks, Control, Sensing, Variables).
3. Students upload their morning DALL·E character art as a new sprite costume (Costumes tab → Upload) and their board/background art as a Stage backdrop.
4. Facilitator live-codes the core loop on the projector (movement + a collectible + score variable) using the exact script below, and students copy it into their own project, then re-theme it (renaming variables, swapping sprites) to match their own concept.
5. Once basic movement + scoring works, introduce ElevenLabs for sound effects. Model one generation live, download the .mp3, and show how to import it into a Scratch sprite's Sounds tab.
6. Students generate 2–3 sound effects matching their game's key moments (collect, win, lose) and attach them to the matching Scratch events.

7. After the break, students add a background, a "You Win"/"Game Over" message using broadcast messages, and a visible score display.
8. Run a structured playtest rotation: pair students up, each plays the other's game for 3 minutes, then fills a feedback index card using 3 sentence starters: "I liked...", "I was confused by...", "I would suggest...".
9. Students spend 10 minutes making one specific, feedback-driven change (e.g. slowing movement speed, adding an instructions screen, fixing a broken collision).
10. Close with a gallery walk: laptops open on desks, students circulate and play 2–3 other games, then everyone returns to seats for shout-outs.

Build detail: AI tools, workflow, and exact prompts

Tool setup: Scratch at scratch.mit.edu (or the offline editor if internet is unreliable). ElevenLabs Sound Effects at elevenlabs.io, free account, using the "Sound Effects" generator tool (text-to-sound-effect, not the voice cloning tool).

Step 1 — Core Scratch script. Facilitator live-codes this exact block sequence on the Player sprite (students rename sprite/variable labels to match their theme, e.g. "Zawadi" instead of "Player", "Snares" instead of "Score"):

```
when green flag clicked
set [Score v] to 0
go to x: (-200) y: (-100)
forever
  if  then
    change x by (5)
  end
  if  then
    change x by (-5)
  end
  if  then
    change y by (5)
  end
  if  then
    change y by (-5)
  end
  if  then
    change [Score v] by (1)
    play sound [collect-sfx v]
    go to random position
  end
  if <(Score) = (10)> then
    broadcast [You Win v]
    stop [this script v]
  end
end
```

On the Collectible sprite:

```
when green flag clicked
forever
  go to random position
end
```

On the Stage, to display the win screen:

```
when I receive [You Win v]
switch backdrop to [win-screen v]
play sound [win-sfx v]
```

Good result looks like: the sprite moves smoothly with arrow keys, the score visibly increases when touching the collectible, the collectible relocates, and reaching the target score triggers a clear win state with sound. Students test by playing their own game 3+ times, checking that (a) controls feel responsive, not too fast/slow, and (b) the win condition actually triggers.

Step 2 — AI sound effect generation (ElevenLabs). Example prompts to type into the Sound Effects generator:

```
Short, cheerful 8-bit style "coin collect" chime, under 1 second

Soft footsteps running on dry grass, looping, 2 seconds

Triumphant short victory fanfare, retro game style, 2-3 seconds

Sad descending "game over" tone, short, retro game style
```

Workflow: generate 2–3 variations per prompt, preview each, download the best as an .mp3, then in Scratch go to the relevant sprite's Sounds tab → Upload Sound → select the file → rename it clearly (e.g. "collect-sfx") so it matches the block script. Good result: a short (under 3 second) sound that clearly matches the in-game moment and isn't jarringly loud compared to others — students should preview all sounds together and normalise choices (e.g. don't mix a very loud fanfare with a very quiet chime).

Step 3 — Playtest feedback capture. Give students this exact structure on an index card:

```
PLAYTEST FEEDBACK CARD
Game title: ___________________
Tester name: ___________________
```

```
I liked: ___________________
I was confused by: ___________________
I would suggest: ___________________
```

Students must action at least one "I would suggest" comment before the showcase — this is a graded requirement (see Assessment section).

Checks for understanding

- "Which block makes your score go up, and where is it in your script?"
- "What happens in your game when the win condition is met — walk me through it."
- "What's one thing your playtester was confused by, and what did you change because of it?"

Common mistakes & fixes

MISTAKE	FIX
Sprite moves off-screen and gets lost	Add "if on edge, bounce" block or an "x position between -240 and 240" check
Score doesn't increase (collision not detected)	Check both sprites' exact names match in the "touching" block dropdown; resize sprite hitboxes if art has excess transparent padding
Sound doesn't play or is too quiet/loud	Re-check the sound is uploaded to the correct sprite, not the Stage; use "set volume" block to balance levels across sounds
Win condition never triggers	Print the Score variable on stage (checkbox in Variables palette) so students can see it updating live and debug the threshold value
Student ignores playtest feedback and changes nothing	Facilitator requires one specific, named change ("what will you change because of X's comment?") before sign-off

Assessment & Showcase

Assessment is formative and portfolio-based rather than test-based — the goal is a genuine creative artefact, not a quiz score. The facilitator circulates throughout both sessions using a simple 3-level check (Not yet / Getting there / Done) against the following "done" checklist, and records it on a class tracking sheet:

- **Design portfolio** contains: concept + pitch, full written rules, at least one character backstory, at least one AI-generated character image and one board/

background image, and a short written reflection on one change made after playtesting.
- **Scratch game** has: working player movement, a functioning score variable, at least one AI-generated sound effect correctly triggered, and a clear win condition (and ideally a lose condition).
- **Evidence of iteration**: at least one specific change made in direct response to peer playtest feedback, named in the portfolio reflection.

The final 10 minutes of the afternoon session run as a **gallery walk showcase**: all laptops stay open on desks with games ready to play, students circulate in small groups playing 2–3 classmates' games, and the facilitator invites 2–3 volunteers to demo their game to the whole room, sharing one design decision and one thing they changed after feedback. Students then export/submit their portfolio PDF (File → Download/Export as PDF from the shared template) and share their Scratch project link (or export the .sb3 file to USB) before leaving — this is the formal "done" moment and the trigger for handing over take-home materials.

Extension & Home Challenges

- **Easy:** Add a second collectible type worth different points, and generate one new AI sound effect and one new character image for it. Update the portfolio with the new addition.
- **Medium:** Add a two-player mode (second sprite controlled with WASD keys) or a simple timer/lives system using a new Scratch variable, and use ChatGPT to write an updated rules section covering the new mechanic.
- **Hard:** Build a branching mini-story using Scratch's "ask and wait" block and a "list" variable to track player choices, generate 2–3 alternate character portraits for different story paths with Bing Image Creator, and write a full "Game Design Document" in ChatGPT (concept, mechanics, art direction, sound direction) formatted as a shareable one-pager for the portfolio.